

# PREDICTIVE ANALYSIS OF RURAL WATER PUMP FUNCTIONALITY

Analyzing Tanzania's Water Pump Systems Through ML and  
Interactive Dashboards

**Team 5:**  
**Haruka Inoba**  
**Tetiana Bogach**  
**Anandhu Krishnan Anitha Vijayan**

We use tech to connect human potential and  
opportunity with dignity & humility

**PROJECT**

**OVERVIE**

**W**



# ***Predictive Analysis of Rural Water Pump Functionality***

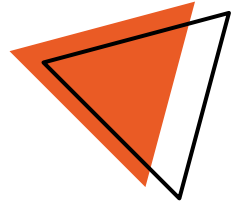


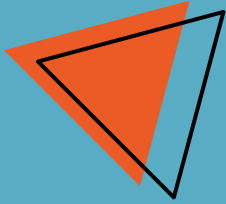
**Problem:** Sustainable water service delivery is a major challenge in rural Tanzania, where approximately **45% of water points are non-functional** (as of a 2013 survey). Reactive maintenance is costly and inefficient.

**Solution:** Developed a complete Data Intelligence Pipeline integrating Exploratory Data Analysis (EDA), **Geospatial Analytics** and Machine Learning.

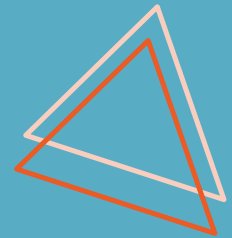
**Dataset:** Analyzed **59,400+** water pump installations across Tanzania using features like geographic coordinates, construction year, water source, and management structure.

**Objective:** Build a model that predicts whether a pump is functional, non-functional, or requires repair, supporting data-driven resource allocation.





***The final output is an interactive Streamlit Dashboard designed as a Decision-Support Tool to help NGOs, the Tanzanian Ministry of Water, and local Community-Based Water Organizations (CBWSOs) prioritize maintenance efforts and allocate scarce resources effectively.***

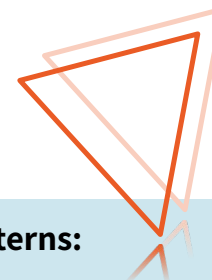


Two decorative orange circles on the left side of the slide. The top one is a solid orange circle with a thin white outline, and the bottom one is a solid orange circle with a thin white outline.

# EXPLORATORY DATA INSIGHTS



# Key Analytical & Geospatial Findings



## Geospatial Clustering:

- Functional Pumps cluster in Northern regions (e.g., Mwanza/Shinyanga) with higher altitude (gps\_height) and better-structured management schemes.
- Non-Functional Pumps concentrate in remote basins and regions (e.g., Dodoma/Mtwara) often characterized by low elevation and non-existent user payment systems.



## Operational & Financial Insights:

- Payment Type is a critical differentiator. Non-functional pumps are highly concentrated in the 'never pay' category, highlighting the critical link between revenue collection and sustained service delivery.
- Pump Age shows a clear non-linear failure rate, with a sharp increase in failure probability after the 20–25 year mark.



## Temporal Patterns:

- Interactive analysis reveals spikes in pump breakdowns coinciding with historical drought periods, suggesting the importance of incorporating external weather data.





# MACHINE LEARNING MODEL



# ***Model Architecture & Performance***

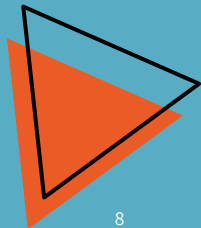
## ***Model Inputs***

### ***Categorical***

- ***Quantity Group***
- ***Region***
- ***Extraction Type Class***
- ***Source***
- ***Basin***
- ***Payment\_type***

### ***Numerical***

- ***Pump Age***
- ***GPS Height***



# ***Model Characteristics***



*Our machine learning model is specifically designed to be lightweight and highly interpretable, making it suitable for operational deployment in resource-constrained environments.*





**75.0%**

***Training Accuracy***

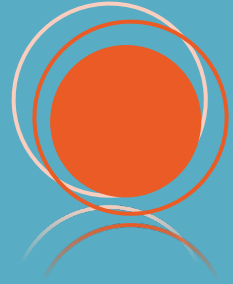
***Achieved during  
model development***

**73.0%**

***Test Accuracy***

***Demonstrates stable  
generalization to new  
data***



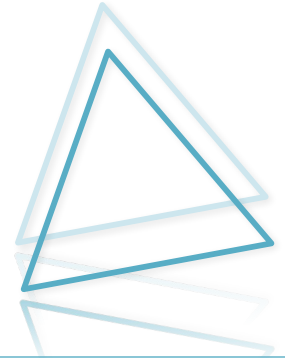
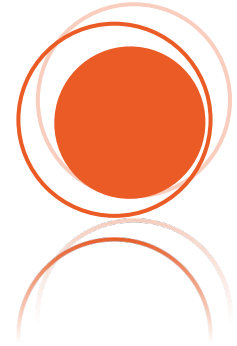


- ***Feature importance analysis revealed that pump\_age, extraction\_type\_class, and payment\_type are the leading predictors of pump functionality.***
- ***The model was deliberately optimized for interpretability over raw accuracy, ensuring field engineers can trust and effectively act upon its predictions.***





# STREAMLIT DASHBOARD



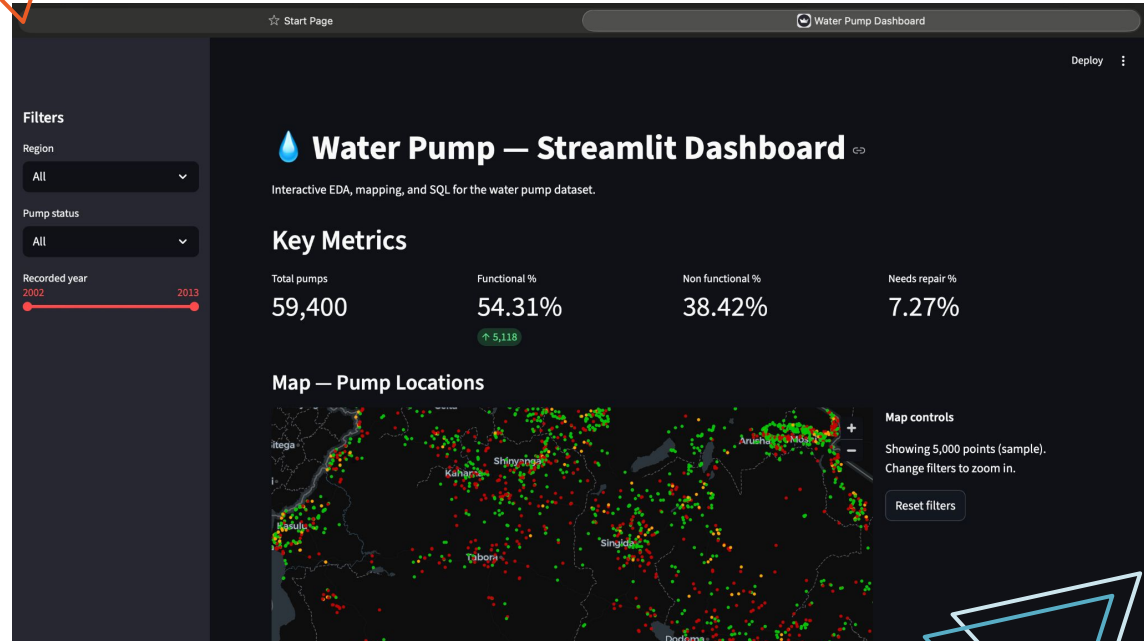
# Interactive Analytical Platform

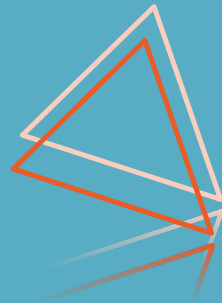
**KPI Summaries:** At-a-glance cards presenting total pumps, condition percentages, and year-based trends.

**Geospatial Visualisation:** Real-time mapping with zoom, filtering, and reset features for detailed location analysis.

**SQL Query Interface:** A built-in interface allows analysts fine-grained control for executing complex data queries.

**Data Triage Support:** Enables rapid scenario-based filtering, such as identifying pumps needing repair installed before 2005 in specific basins.

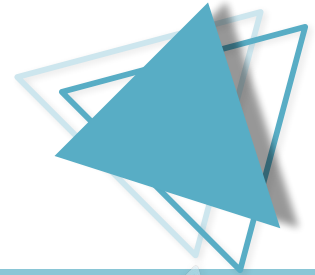




***Dashboard architecture designed for plug-and-play model upgrades,  
allowing future ML versions to be deployed without UI changes.***



# FUTURE ENHANCEMENTS & STRATEGIC OUTLOOK



# ***Scalability, Impact & Next Steps***

## **Technical Improvements**

- Replace coarse geographical categories with numerical coordinates for more fine-grained splits in tree models, enhancing predictive precision.
- Evaluate advanced machine learning models such as XGBoost, CatBoost, LightGBM, and spatial deep learning for improved performance.
- Incorporate additional environmental data, including rainfall, soil type, and distance-from-nearest-town, to enrich contextual analysis.

## **Deployment & Real-World Impact**

- Transform the dashboard into a comprehensive maintenance decision engine, offering proactive recommendations for repair priorities.
- Integrate mobile data collection, allowing field engineers to update pump conditions directly from the field in real-time.
- Automate alerts for high-risk pumps identified through model forecasting, ensuring timely intervention and preventing failures.



# Thanks a lot!



***Grateful for the guidance and support that shaped this project into something meaningful.***

***Thank you for helping me grow into a more confident and capable data professional.***

